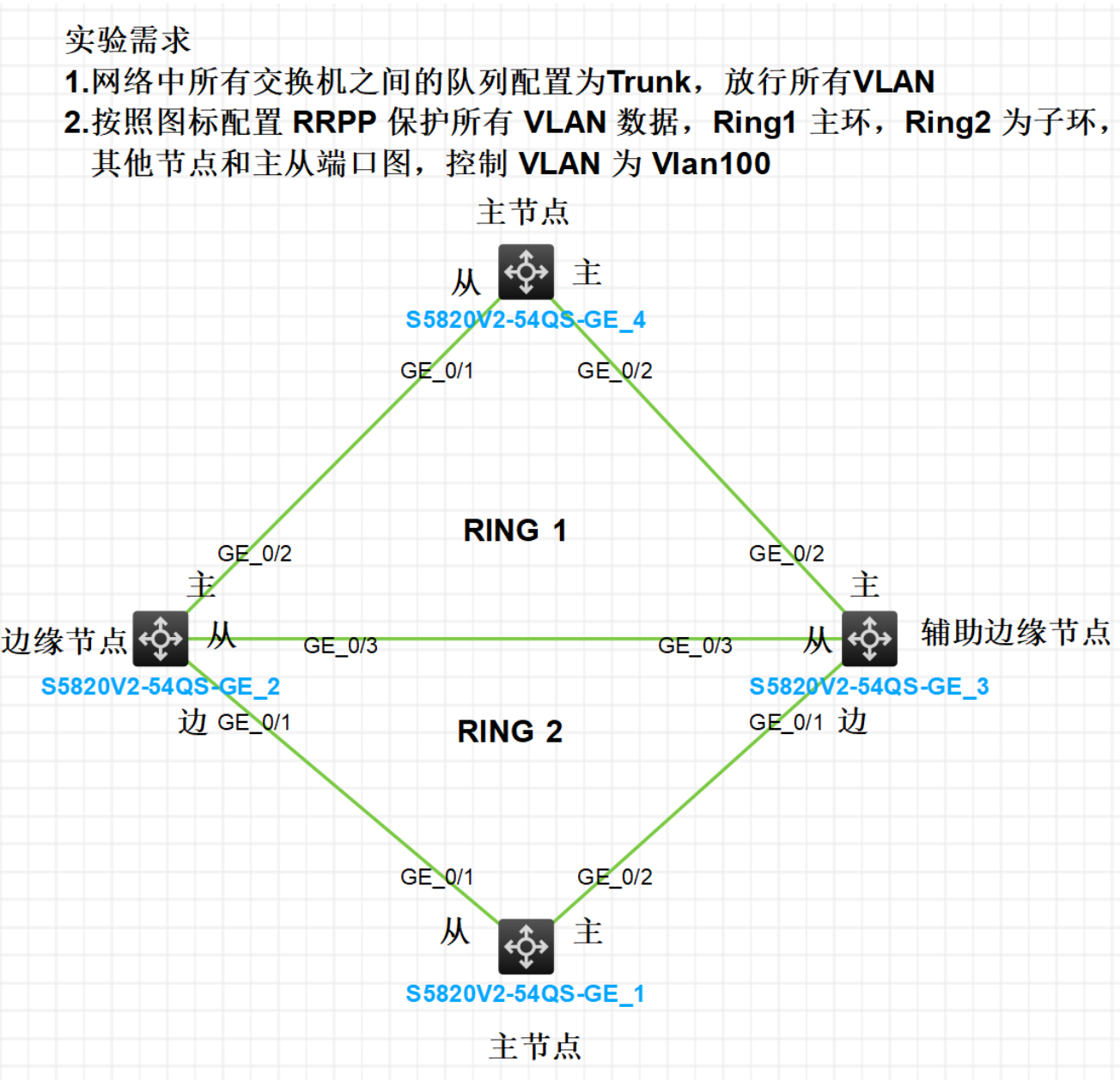


RRPP实验

实验拓扑



实验需求

1. 网络中所有交换机之间的队列配置为Trunk，放行所有VLAN
2. 按照图标配置RRPP保护所有VLAN数据，Ring1为主环，Ring2为子环，其他节点和主从端口图，控制VLAN为Vlan100

实验配置

```

[SW1] vlan 100
[SW1] int ra g 1/0/1 to g 1/0/2
[SW1-if-range] port link-type trunk
[SW1-if-range] port trunk permit vlan all
[SW1-if-range] undo stp enable
[SW1] rrpp do 1
[SW1-rrpp-domain1] control-vlan 4092
[SW1-rrpp-domain1] protected-vlan reference-instance 0 to 32
[SW1-rrpp-domain1] ring 2 node-mode master primary-port g1/0/2 secondary-port Gi
gabitEthernet 1/0/1 level 1
[SW1-rrpp-domain1] ring 2 enable
[SW1] rrpp enable
---
[SW2] vlan 4092
[SW2-vlan4092] vlan 4093
[SW2] int ra g 1/0/1 to g 1/0/3
[SW2-if-range] port link-type trunk
[SW2-if-range] port trunk permit vlan all
[SW2-if-range] undo stp enable
[SW2] rrpp do 1
[SW2-rrpp-domain1] control-vlan 100
[SW2-rrpp-domain1] protected-vlan reference-instance 0 to 32
[SW2-rrpp-domain1] ring 1 node-mode transit primary-port g1/0/2 secondary-port Gi
gabitEthernet 1/0/3 level 0
[SW2-rrpp-domain1] ring 2 node-mode edge edge-port g1/0/1
[SW2-rrpp-domain1] ring 1 enable
[SW2-rrpp-domain1] ring 2 enable
[SW2] rrpp enable
---
[SW3] vlan 4092
[SW3-vlan4092] vlan 4093
[SW3] int ra g 1/0/1 to g 1/0/3
[SW3-if-range] port link-type trunk
[SW3-if-range] port trunk permit vlan all
[SW3-if-range] undo stp enable
[SW3] rrpp do 1
[SW3-rrpp-domain1] control-vlan 100
[SW3-rrpp-domain1] protected-vlan reference-instance 0 to 32
[SW3-rrpp-domain1] ring 1 node-mode transit primary-port g1/0/2 secondary-port Gi
gabitEthernet 1/0/3 level 0
[SW3-rrpp-domain1] ring 2 node-mode assistant-edge edge-port g1/0/1
[SW3-rrpp-domain1] ring 1 enable
[SW3-rrpp-domain1] ring 2 enable
[SW3] rrpp enable
---
[SW4] vlan 4092
[SW4-vlan4092] vlan 4093
[SW4] int ra g 1/0/1 to g 1/0/2
[SW4-if-range] port link-type trunk
[SW4-if-range] port trunk permit vlan all
[SW4-if-range] undo stp enable
[SW4] rrpp domain 1
[SW4-rrpp-domain1] control-vlan 100

```

```
[SW4-rrpp-domain1]protected-vlan reference-instance 0 to 32
[SW4-rrpp-domain1]ring 1 node-mode master primary-port g1/0/2 secondary-port g1/
0/1 level 0
[SW4-rrpp-domain1]ring 1 enable
[SW4]rrpp enable
```

结果验证

```
<SW1>display rrpp brief
Flags for node mode: M -- Master, T -- Transit, E -- Edge, A -- Assistant-edge

RRPP protocol status: Enabled

Domain ID      : 1
Control VLAN   : Primary 4092, Secondary 4093
Protected VLAN: Reference instance 0 to 32
Hello timer    : 1 seconds, Fail timer: 3 seconds
Fast detection status: Disabled
Fast-Hello timer: 20 ms, Fast-Fail timer: 60 ms
Fast-Edge-Hello timer: 10 ms, Fast-Edge-Fail timer: 30 ms
Ring ID        : 2
Ring level     : 1
Node mode      : Master
Ring state     : Complete
Enable status  : Yes, Active status: Yes
Primary port   : GE1/0/2
Secondary port : GE1/0/1

-----
 2   1   M   GE1/0/2           GE1/0/1           Yes
-----

<SW1>display rrpp verbose domain 1
Domain ID      : 1
Control VLAN   : Primary 4092, Secondary 4093
Protected VLAN: Reference instance 0 to 32
Hello timer    : 1 seconds, Fail timer: 3 seconds
Fast detection status: Disabled
Fast-Hello timer: 20 ms, Fast-Fail timer: 60 ms
Fast-Edge-Hello timer: 10 ms, Fast-Edge-Fail timer: 30 ms

Ring ID        : 2
Ring level     : 1
Node mode      : Master
Ring state     : Complete
Enable status  : Yes, Active status: Yes
Primary port   : GE1/0/2
Secondary port : GE1/0/1
Port status: UP
Port status: BLOCKED

---

<SW2>display rrpp brief
Flags for node mode: M -- Master, T -- Transit, E -- Edge, A -- Assistant-edge

RRPP protocol status: Enabled

Domain ID      : 1
Control VLAN   : Primary 100, Secondary 101
Protected VLAN: Reference instance 0 to 32
Hello timer    : 1 seconds, Fail timer: 3 seconds
```

```
Fast detection status: Disabled
Fast-Hello timer: 20 ms, Fast-Fail timer: 60 ms
Fast-Edge-Hello timer: 10 ms, Fast-Edge-Fail timer: 30 ms
```

Ring ID	Ring level	Node mode	Primary/Common port	Secondary/Edge port	Enable status
1	0	T	GE1/0/2	GE1/0/3	Yes
2	1	E	GE1/0/2 GE1/0/3	GE1/0/1	Yes

```
<SW2>display rrpp verbose domain 1
Domain ID      : 1
Control VLAN   : Primary 100, Secondary 101
Protected VLAN : Reference instance 0 to 32
Hello timer    : 1 seconds, Fail timer: 3 seconds
Fast detection status: Disabled
Fast-Hello timer: 20 ms, Fast-Fail timer: 60 ms
Fast-Edge-Hello timer: 10 ms, Fast-Edge-Fail timer: 30 ms
```

```
Ring ID      : 1
Ring level   : 0
Node mode    : Transit
Ring state   : -
Enable status : Yes, Active status: Yes
Primary port  : GE1/0/2          Port status: UP
Secondary port: GE1/0/3          Port status: UP
```

```
Ring ID      : 2
Ring level   : 1
Node mode    : Edge
Ring state   : -
Enable status : Yes, Active status: Yes
Common port  : GE1/0/2          Port status: UP
              GE1/0/3          Port status: UP
Edge port    : GE1/0/1          Port status: UP
```

```
---
<SW3>display rrpp brief
Flags for node mode: M -- Master, T -- Transit, E -- Edge, A -- Assistant-edge

RRPP protocol status: Enabled
```

```
Domain ID      : 1
Control VLAN   : Primary 100, Secondary 101
Protected VLAN : Reference instance 0 to 32
Hello timer    : 1 seconds, Fail timer: 3 seconds
Fast detection status: Disabled
Fast-Hello timer: 20 ms, Fast-Fail timer: 60 ms
Fast-Edge-Hello timer: 10 ms, Fast-Edge-Fail timer: 30 ms
```

Ring ID	Ring level	Node mode	Primary/Common port	Secondary/Edge port	Enable status
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1	0	T	GE1/0/2	GE1/0/3	Yes
2	1	A	GE1/0/2	GE1/0/1	Yes
			GE1/0/3		

```
<SW3>display rrpp verbose domain 1
Domain ID      : 1
Control VLAN   : Primary 100, Secondary 101
Protected VLAN: Reference instance 0 to 32
Hello timer    : 1 seconds, Fail timer: 3 seconds
Fast detection status: Disabled
Fast-Hello timer: 20 ms, Fast-Fail timer: 60 ms
Fast-Edge-Hello timer: 10 ms, Fast-Edge-Fail timer: 30 ms
```

```
Ring ID      : 1
Ring level   : 0
Node mode    : Transit
Ring state   : -
Enable status : Yes, Active status: Yes
Primary port  : GE1/0/2          Port status: UP
Secondary port: GE1/0/3          Port status: UP
```

```
Ring ID      : 2
Ring level   : 1
Node mode    : Assistant-edge
Ring state   : -
Enable status : Yes, Active status: Yes
Common port  : GE1/0/2          Port status: UP
              GE1/0/3          Port status: UP
Edge port    : GE1/0/1          Port status: UP
```

```
---
<SW4>display rrpp brief
Flags for node mode: M -- Master, T -- Transit, E -- Edge, A -- Assistant-edge
```

```
RRPP protocol status: Enabled
```

```
Domain ID      : 1
Control VLAN   : Primary 100, Secondary 101
Protected VLAN: Reference instance 0 to 32
Hello timer    : 1 seconds, Fail timer: 3 seconds
Fast detection status: Disabled
Fast-Hello timer: 20 ms, Fast-Fail timer: 60 ms
Fast-Edge-Hello timer: 10 ms, Fast-Edge-Fail timer: 30 ms
```

Ring ID	Ring level	Node mode	Primary/Common port	Secondary/Edge port	Enable status
1	0	M	GE1/0/2	GE1/0/1	Yes

```
<SW4>display rrpp verbose domain 1
Domain ID      : 1
Control VLAN   : Primary 100, Secondary 101

Protected VLAN: Reference instance 0 to 32
```

```
Hello timer    : 1 seconds, Fail timer: 3 seconds
Fast detection status: Disabled
Fast-Hello timer: 20 ms, Fast-Fail timer: 60 ms
Fast-Edge-Hello timer: 10 ms, Fast-Edge-Fail timer: 30 ms

Ring ID       : 1
Ring level    : 0
Node mode     : Master
Ring state    : Complete
Enable status : Yes, Active status: Yes
Primary port  : GE1/0/2          Port status: UP
Secondary port: GE1/0/1          Port status: BLOCKED
```

附录

- 1.在前次实验中需创建VLAN 100、VLAN 4092等，否则状态会停留在Failed状态；而在此次重新配置了该实验却不需配置VLAN
 - 2.在第二次中发现第一个问题仍未解决，经查资料发现是"用来承载 RRPP 报文的 控制 VLAN 要不能在二层上正常二层转发，没有创建控制VLAN可能导致被丢包，所以RRPP为failed状态
- 解决：在所有交换机上创建控制VLAN